Data Cleaning Guide

E-Commerce Sales Drop Analysis

Important: these are synthetic portfolio-learning datasets, not real company data. Treat hypotheses as questions to test, not established findings.

Purpose

This guide describes the intended, reproducible transformation from the supplied raw CSVs to the supplied cleaned CSVs. The cleaned files are reference outputs. The rules below should be followed in order, with row counts and key uniqueness checked after each stage.

Column removal policy: no original source columns are removed in these projects. Derived columns are added where useful. If you choose a different policy, document the reason and treat it as a deliberate variation.

File-level cleaning plan

Raw file	Raw rows / cols	Clean file	Clean rows / cols	Main expected action
customers_raw.csv	6,518 / 7	customers_clean.csv	6500	Deduplicate, normalize types/labels, derive analytical fields where documented
orders_raw.csv	32,096 / 11	orders_clean.csv	32000	Deduplicate, normalize types/labels, derive analytical fields where documented
products_raw.csv	68 / 6	products_clean.csv	65	Deduplicate, normalize types/labels, derive analytical fields where documented

Detailed transformations

File / column(s)	Problem in raw data	Transformation / decision	Expected output
orders_raw.csv	96 duplicate rows appended; 32,000 unique order records expected after exact-row deduplication. Currency symbols in unit_price and revenue; mixed order_date formats; inconsistent sales_channel casing/whitespace; order_status whitespace/casing; some blank customer_id values.	Drop exact duplicate rows; parse dates to YYYY-MM-DD; strip ■ and commas then convert unit_price/revenue to numeric; trim and title-case sales_channel and order_status; retain blank customer_id rather than inventing an identity; derive month, gross_sales, discount_amount, profit_estimate.	No source columns removed. Four derived columns added: month, gross_sales, discount_amount, profit_estimate.
customers_raw.csv	18 duplicate rows; city and customer_segment inconsistent casing; 30 signup dates in day-month-year format.	Deduplicate by customer_id; standardize city/region/segment labels; parse signup_date to YYYY-MM-DD.	No columns removed.
products_raw.csv	3 duplicate rows; category casing inconsistent; 4 list_price values formatted with ■.	Deduplicate by product_id; standardize category; remove currency symbol and convert list_price to numeric.	No columns removed.

Validation checklist:

• Check the raw and cleaned row counts against the table above.

• Check the documented primary key is unique after deduplication.

• Check dates parse successfully and use the expected standard format.

• Check numeric fields are numeric and contain no currency symbols.

• Check category/event labels have consistent casing and whitespace.

• Reconcile derived metrics against their formulas.

• Do not delete missing identifiers or impute values unless the rule above explicitly says so.

Column-level notes

No original columns are removed in this project. Derived fields and their formulas are documented below.

• orders.month = month portion of order_date in YYYY-MM format.

• orders.gross_sales = unit_price × quantity.

- orders.discount_amount = gross_sales × discount_pct.
- orders.profit_estimate = revenue − estimated_cost.
- Blank customer_id values are retained; do not invent customer identities.

Reproducibility note

CSV tools may infer types differently (for example, customer IDs should remain text, not numbers). Keep IDs as text, preserve leading zeros, and use the same duplicate key specified in the guide.

If your cleaned output differs from the reference file, first compare parsing rules, duplicate logic, missing-value handling, and type conversions. A difference is not automatically an error, but it must be explained.